

# Requirements Specification

## Leaf & Root E-Commerce Website

### DG4TPR Software Development Project

**Version:** 1.0

**Date:** July 2026

**Project Team:** Leaf & Root Development Team

## 1. Introduction

### 1.1 Purpose

The purpose of this project is to design, develop and deploy an e-commerce website for **Leaf & Root**, a business specialising in the sale of houseplants, succulents, cacti, plant accessories and related products.

The website will allow customers to browse products, search for items, add products to a basket and complete purchases online. The website will also provide administrative functionality to manage customer orders and inventory.

### 1.2 Project Objectives

The objectives of the system are:

- Provide an online platform for selling products.
- Allow customers to search and browse products.
- Enable customers to purchase products through an online checkout process.
- Allow administrators to view and manage customer orders.
- Support stock and inventory management.
- Improve customer accessibility and convenience.

## 2. Stakeholders

| Stakeholder      | Role                                                 |
|------------------|------------------------------------------------------|
| Customers        | Purchase products and submit enquiries               |
| Guest Customers  | Purchase products without registering                |
| Administrators   | Manage products, stock and orders                    |
| Business Owner   | Monitors website performance and business operations |
| Development Team | Design, develop and maintain the system              |

## 3. Functional Requirements

### Priority 1 (Must Have)

#### FR1 - Product Catalogue

The system shall provide a products page displaying all products sold by the business.

#### Acceptance Criteria

- Display product name.
- Display product image.
- Display product description.
- Display product price.
- Display category.
- Display stock availability.

#### FR2 - Product Search

The system shall enable customers to search for products by name.

### **Acceptance Criteria**

- Search text box available.
- Partial matches supported.
- Relevant products displayed.

### **FR3 - Category Filtering**

The system shall enable customers to filter products by category.

#### **Categories**

- Indoor Plants
- Succulents
- Cacti
- Pots
- Accessories

### **Acceptance Criteria**

- Products filtered correctly.
- Multiple categories displayed.

### **FR4 - Product Details**

The system shall provide a separate page displaying detailed product information.

### **Acceptance Criteria**

- Product image displayed.
- Product description displayed.
- Product variants displayed.
- Price displayed.
- Stock status displayed.

## **Priority 2 (Must Have)**

### **FR5 - Shopping Basket**

The system shall provide a basket page displaying products selected by the customer.

## **Acceptance Criteria**

Customers shall be able to:

- Add products to basket.
- Remove products from basket.
- Change product quantities.
- View basket totals.

## **FR6 - Basket Validation**

The system shall validate stock availability before products are added to the basket.

## **Acceptance Criteria**

- Prevent unavailable products being purchased.
- Display relevant error messages.

## **Priority 3 (Must Have)**

## **FR7 - Guest Checkout**

The system shall allow customers to complete an order without creating an account.

## **Customer Information**

- Full Name
- Email Address
- Telephone Number
- Delivery Address

## **Acceptance Criteria**

- Order submitted successfully.
- Customer receives confirmation.

## **FR8 - Payment Simulation**

The system shall include a simulated payment process for demonstration purposes.

### **Acceptance Criteria**

- Payment information captured.
- Order generated after payment confirmation.

### **FR9 - Order Creation**

The system shall create an order record following successful checkout.

### **Acceptance Criteria**

- Order stored in database.
- Order items stored.
- Order totals calculated.

## **Priority 4 (Must Have)**

### **FR10 - Admin Authentication**

The system shall provide password-protected administrator access.

### **Acceptance Criteria**

- Admin login page.
- Password validation.
- Restricted access to admin pages.

### **FR11 - Order Management**

The system shall allow administrators to view and process customer orders.

### **Acceptance Criteria**

Administrators can:

- View all orders.
- Search orders.
- Filter orders.
- Update order status.

## **Order Statuses**

- Pending
- Processing
- Shipped
- Delivered

## **Priority 5 (Should Have)**

### **FR12 - Customer Registration**

The system shall allow customers to create an account.

#### **Acceptance Criteria**

- Registration form available.
- User account stored.
- Duplicate email addresses prevented.

## **Priority 6 (Should Have)**

### **FR13 - Customer Login**

The system shall allow registered customers to log in.

#### **Acceptance Criteria**

- Secure authentication.
- Password validation.
- Logout functionality.

### **FR14 - Admin Login Enhancements**

The system shall support role-based access control.

#### **Acceptance Criteria**

- Admin-only pages protected.
- Customers denied admin access.

## **Priority 7 (Should Have)**

### **FR15 - Contact Us**

The system shall provide a contact page.

#### **Acceptance Criteria**

Display:

- Business address
- Telephone number
- Email address

### **FR16 - Contact Form**

Customers shall be able to submit enquiries.

#### **Acceptance Criteria**

- Form validation.
- Message stored in database.
- Message viewable by administrator.

## **Priority 8 (Could Have)**

### **FR17 - About Us Page**

The system shall provide information about the company.

#### **Acceptance Criteria**

Display:

- Business story
- Vision statement
- Mission statement

## **Priority 9 (Could Have)**

### **FR18 - Home Page**

The system shall provide a homepage introducing the business.

#### **Acceptance Criteria**

Display:

- Featured products
- Categories
- Promotions
- Navigation links

## **4. Inventory Management Requirements**

### **FR19 - Product Management**

Administrators shall manage products.

#### **Acceptance Criteria**

- Add products.
- Edit products.
- Delete products.

### **FR20 - Inventory Management**

Administrators shall manage stock levels.

#### **Acceptance Criteria**

- Record stock quantity.
- Update inventory.
- Display low stock warnings.



## **FR21 - Inventory Reporting**

The system shall generate inventory reports.

### **Acceptance Criteria**

Display:

- Current stock.
- Low stock products.
- Out-of-stock products.

# **5. Non-Functional Requirements**

## **NFR1 - Usability**

The system shall:

- Be easy to navigate.
- Use a consistent layout.
- Provide clear feedback to users.

## **NFR2 - Performance**

The system shall:

- Return search results within 3 seconds.
- Load pages efficiently.

## **NFR3 - Security**

The system shall:

- Encrypt passwords.
- Protect admin pages.
- Validate all user input.
- Prevent SQL injection attacks.

## **NFR4 - Reliability**

The system shall:

- Store data accurately.
- Recover gracefully from errors.
- Maintain database integrity.

## **NFR5 - Compatibility**

The website shall function on:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

## **NFR6 - Responsiveness**

The website shall be usable on:

- Desktop devices
- Tablets
- Mobile devices

# **6. MoSCoW Prioritisation**

## **Must Have**

1. Product Catalogue
2. Product Search
3. Category Filtering
4. Product Details
5. Basket
6. Guest Checkout
7. Order Creation
8. Admin Authentication
9. Order Management

## Should Have

1. Customer Registration
2. Customer Login
3. Contact Us
4. Contact Form
5. Inventory Management
6. Inventory Reporting

## Could Have

1. About Us
2. Home Page
3. Reviews
4. Product Recommendations
5. Plant Care Reminders

# 7. Minimum Viable Product (MVP)

The project will be considered submission-ready when the following workflow is operational:

Browse Products



Search Products



Add To Basket



Checkout



Order Created



Admin Views Order

This MVP has been prioritised because it delivers the complete e-commerce transaction process and directly supports the primary business objective of selling products online.

